

A Combinatorial Origin of Locality

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Locality and unitarity are fundamental principles of quantum field theory. For tree-level scattering amplitudes, locality determines which propagator poles may occur together, while unitarity fixes their residues through factorization. Previous uniqueness theorems have repeatedly shown that unitarity can emerge from locality combined with other physical principles. In this Letter we give the first complete all-multiplicity proof for the emergence of locality in this setting. We focus on $\text{Tr}(\phi^3)$ theory and phrase locality as a problem in geometric combinatorics. Planar propagators are chords of a polygon, and a denominator is local exactly when its chords triangulate the polygon, and non-local otherwise. We show that hidden zeros—regular kinematic conditions on which the amplitude vanishes—enforce pole compatibility and therefore imply a local singularity structure. The proof rests on two ingredients: a graphical *star test* determines when a chosen zero eliminates a pole product, and a *smoothing* procedure finds such a zero. Thus, once the planar pole alphabet and denominator bound are specified, locality and unitarity emerge together from a single on-shell principle: hidden zeros. Equivalently, our proof provides a novel characterization of a familiar concept: polygon triangulations are precisely the chord configurations that evade every star test.

INTRODUCTION

The Lagrangian formulation builds quantum field theory from interactions localized in spacetime and a unitary time evolution. At tree level, locality means that the propagators appearing together in a pole product must belong to a single Feynman graph, while unitarity requires factorization on those poles. Modern on-shell and geometric formulations raise a sharper question: must the local graph structure be assumed, or can locality and unitarity emerge from a different starting point [1–4]? Deriving both locality and unitarity would point toward formulations of quantum field theory based on principles different from those of a local Lagrangian.

Many uniqueness results have already shown that unitarity can emerge. Starting from a local pole ansatz, gauge invariance, soft behavior, or ultraviolet scaling can uniquely fix the amplitude and recover its correct factorization [4–9]. Locality remained an input in the all-multiplicity proofs, although finite-multiplicity evidence suggested that it too could emerge [4, 10, 11] (see also related developments [12–16]).

In this Letter we give the first complete all-multiplicity proof that locality can itself emerge from other on-shell properties. To isolate the heart of the problem—the compatibility of propagator poles—we work with color-ordered $\text{Tr}(\phi^3)$ n -point tree amplitudes. Despite its simplicity, this theory retains the full combinatorics of planar poles, while its constant numerators remove the additional structures that complicate the nonlinear sigma model (NLSM) and Yang–Mills theory. We bootstrap it from the minimal data needed for a finite ansatz: the planar pole alphabet, arbitrary constant coefficients, and a denominator bound of at most $n - 3$ factors. We assume neither the exact pole count, simple poles, nor compatibility, and impose

no factorization or residue data.

What remains is to find a bootstrap principle capable of determining both which poles may coexist and their relative coefficients. Hidden zeros [10] (see also [17–27])—regular kinematic conditions on which the amplitude vanishes without requiring a physical propagator to go on shell—provide precisely such a principle. Although defined without reference to poles or factorization, hidden zeros fix amplitudes within a local ansatz and constrain nonlocal terms at finite multiplicity [9–11]. Here we prove this second statement at all multiplicities and uncover the concrete mechanism by which locality emerges.

For a fixed cyclic ordering, each planar channel

$$X_{ij} = (p_i + p_{i+1} + \cdots + p_{j-1})^2$$

is represented by a chord joining vertices i and j of an n -gon. The $n - 3$ propagators of a planar cubic tree triangulate the polygon, and every triangulation is dual to one such tree. A pole product is therefore local precisely when its chords form a triangulation. Motivated by surfaceology [10, 28, 29], we recast the emergence of locality as a problem in geometric combinatorics. This simplest setting exposes the fundamental mechanism cleanly and provides a starting point for the NLSM and Yang–Mills theory, where nontrivial numerators obscure the same pole-compatibility problem.

We prove that every nontriangulating chord configuration is directly incompatible with at least one hidden zero and, more strongly, give a constructive procedure for finding that zero. A graphical *star test* determines whether a proposed zero eliminates a given pole product, while a recursive *smoothing* of the chord configuration constructs a zero that passes the test.

The zeros therefore do more than fix the coefficients of local diagrams: they select the diagrams themselves and

enforce both the correct pole count and simple poles. This strengthens the conjecture of Ref. [10], which assumed exactly $n - 3$ distinct poles per term. Our theorem allows terms with fewer poles and repeated poles and eliminates every such term constructively. Together with the earlier uniqueness theorem [9], this fixes the local, factorizing amplitude up to normalization. Thus, assuming the pole alphabet, locality and unitarity both emerge from hidden zeros.

Read in the opposite direction, the theorem also gives a new characterization of a familiar combinatorial object. Among configurations of at most $n - 3$ chords, polygon triangulations are precisely those that evade every interval star test.

A FIVE-POINT OBSERVATION

We work in massless $\text{Tr}(\phi^3)$ theory, a matrix-valued scalar theory with a cubic single-trace interaction. Fixing the trace order $(1, 2, \dots, n)$ selects a color-ordered n -point tree amplitude whose Feynman diagrams are planar. Their numerators are constant, so this theory isolates the question of which propagator poles may occur together.

For outgoing momenta satisfying $\sum_i p_i = 0$, $p_i^2 = 0$, the possible planar channels are

$$X_{ij} = (p_i + p_{i+1} + \dots + p_{j-1})^2, \quad X_{i,i} = X_{i,i+1} = 0, \quad (1)$$

with labels understood cyclically and complementary channels identified.

At five points there are five channels $\{X_{13}, X_{24}, X_{35}, X_{14}, X_{25}\}$. Allowing at most two denominator factors, the most general constant-coefficient ansatz can be written cyclically as

$$\mathcal{B}_5 = A_0 + \sum_{i=1}^5 \left(\frac{b_i}{X_{i,i+2}} + \frac{r_i}{X_{i,i+2}^2} + \frac{\alpha_i}{X_{i,i+2}X_{i+1,i+3}} + \frac{c_i}{X_{i,i+2}X_{i+2,i+4}} \right). \quad (2)$$

The c_i -terms are the five pole products appearing in planar cubic trees. The remaining terms are the non-local ones. They have too few poles, repeated poles, or pairs of poles that cannot occur together in one such tree.

Now impose the hidden zero (the general zero locus is given in the next section)

$$H_{45} : \quad X_{35} = -X_{14}, \quad X_{25} = X_{24} - X_{14}. \quad (3)$$

This is a regular kinematic locus: X_{13}, X_{24} , and X_{14} remain generic, and no propagator is required to go on shell. Substituting Eq. (3) into Eq. (2) and requiring the result to vanish identically gives, among other conditions, $\alpha_1 = 0$ and $c_1 = c_4$. Thus the zero directly eliminates some nonlocal terms while relating the coefficients of the remaining terms.

Imposing its five cyclic images gives

$$A_0 = b_i = r_i = \alpha_i = 0, \quad c_1 = c_2 = \dots = c_5, \quad (4)$$

and therefore

$$\mathcal{B}_5 = c \sum_{i=1}^5 \frac{1}{X_{i,i+2}X_{i+2,i+4}}, \quad (5)$$

which is the planar five-point amplitude up to an overall normalization.

This example separates the two effects of hidden zeros. They eliminate forbidden denominators and relate the coefficients of the surviving triangulations. It also poses the all-multiplicity problem:

Given a pole product that belongs to no planar cubic tree, which hidden zero eliminates it?

At five points this can be answered by direct substitution. To solve the problem for arbitrary n , we now reformulate it geometrically.

THE CHORD FORMULATION OF LOCALITY

Place the labels $1, \dots, n$ cyclically on the boundary of a disk and represent each nontrivial channel X_{ij} by the chord joining vertices i and j . This is the tree-level disk of surfaceology; the same chord combinatorics underlies the associahedron and dihedral coordinates [3, 30, 31].

A planar cubic tree contains $n - 3$ internal propagators. Its corresponding chords are distinct, do not cross, and triangulate the polygon. Conversely, the graph dual to every triangulation is a planar cubic tree. Thus, within the cubic-tree ansatz, locality has a purely geometric formulation: a pole product is local precisely when its chords form a triangulation. Figure 1 illustrates this correspondence at five points.

At general multiplicity, let M be a multiset of chords and consider

$$\mathcal{B}_n = \sum_{\substack{M \text{ multiset of chords} \\ |M| \leq n-3}} \frac{a_M}{X_M}, \quad X_M = \prod_{ij \in M} X_{ij}, \quad S = \text{supp } M. \quad (6)$$

Using a multiset allows repeated poles, while the bound permits terms with fewer than $n - 3$ factors. We call M *local* when it consists of exactly $n - 3$ distinct chords forming a triangulation, and *nonlocal* otherwise. We call the chords in S selected.

The hidden zeros are labelled by consecutive boundary intervals

$$T = \{w, w + 1, \dots, w + k\}, \quad 2 \leq |T| \leq n - 3. \quad (7)$$

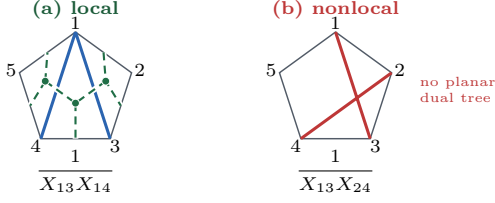


FIG. 1. Five-point locality. Solid chords denote propagators. In (a), the green dashed graph drawn over the triangulation is its dual planar cubic tree. In (b), the crossing chords admit no planar dual.

We call T the *zero interval*. Writing $v = w + k + 1$, the corresponding locus is

$$H_T: \quad X_{w+i,j} = X_{w,j} - X_{w,v} + X_{w+i,v}, \quad (8)$$

$$1 \leq i \leq k, \quad j \in T^c.$$

At a generic point of H_T , no propagator channel is required to vanish. We impose

$$\mathcal{B}_n|_{H_T} = 0 \quad \text{for every zero interval } T. \quad (9)$$

The problem is now entirely geometric. Given a chord multiset that does not triangulate the polygon, find a boundary interval whose zero isolates its coefficient. For $n \geq 5$, we prove that such an interval always exists. Consequently, imposing all hidden zeros eliminates every nonlocal term and leaves only the denominators of planar cubic trees. More strongly, for each nonlocal pole product M , we construct a zero interval T that isolates a_M and forces it to vanish. A separate question is to ask for the minimal number of zeros that accomplishes this fact. We leave this question for future work.

PROOF STRATEGY: WHY CROSSINGS ARE A RED HERRING

A crossing pair makes nonlocality immediately visible, but it turns out a zero cannot detect the crossing of chords itself. Instead, it detects a gap. A region of the polygon that has not been correctly triangulated. At the full cubic pole count, placing a chord in an incompatible position necessarily leaves a triangulating chord missing elsewhere. A crossing then is useful only insofar it signals a chord is missing from its rightful position. So in principle, we need only a geometric mechanism by which a zero detects a non-triangulated region.

What makes this even more difficult is that other misplaced chords can obscure this gap from a particular zero, making the region appear triangulated even though the full configuration is not, as illustrated in Fig. 2. The central task is therefore to prove that $n - 3$ chords are not enough to hide every gap from every interval zero. At least one zero must still expose the nonlocality. Consequently, the successful zero depends on the complete

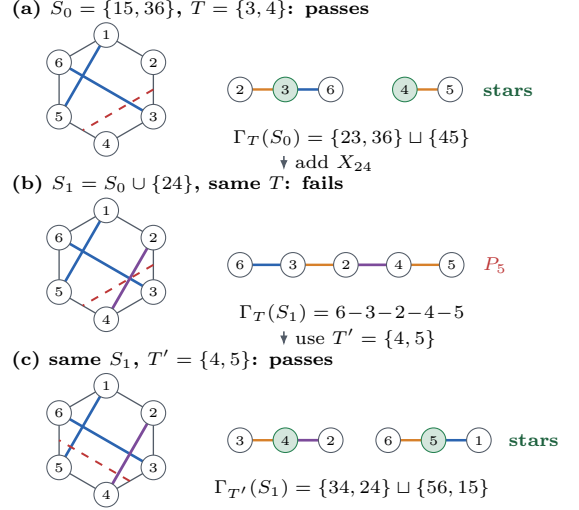


FIG. 2. A misplaced chord can hide a gap from one zero. The cut T passes for S_0 , but adding the crossing chord X_{24} joins its two stars into a P_5 . The enlarged support S_1 is detected by the different cut T' . Red dashes mark the cut; blue and purple links are selected chords, and orange links are its boundary links.

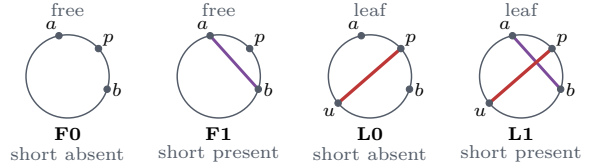


FIG. 3. The four local configurations. The vertices a, p, b are consecutive. In F0 and F1, p is free. In L0 and L1, the red chord X_{pu} is the unique selected chord ending at p . Purple denotes the short chord X_{ab} ; all other chords are suppressed.

chord configuration and cannot, in general, be found by looking only at an obvious crossing pair. This is what makes the proof more non-trivial than one might naively expect given its simple statement.

We solve this problem in two steps. A graphical *star test* first decides whether a proposed hidden zero isolates the chosen denominator. We then find a zero passing this test by recursively removing one simple vertex (free vertex or leaf vertex). Only the four local configurations in Fig. 3 can occur. The main text gives the logic of the recursion; the explicit cuts, lifts, and boundary cases are collected in Appendix 2.

ZEROS AND STARS AS GRAPHICAL DETECTORS

Given a pole product and a candidate hidden zero, we need to determine whether the zero isolates—and therefore eliminates—that term, without substituting it into the full ansatz. To make the mechanism transparent,

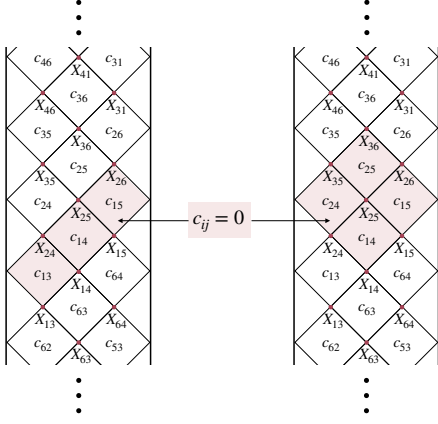


FIG. 4. Kinematic-mesh origin of the star test. The interval $T = \{1, 2\}$ sets $c_{13} = c_{14} = c_{15} = 0$, while $T = \{1, 2, 3\}$ sets $c_{14} = c_{15} = c_{24} = c_{25} = 0$; the corresponding blocks are shaded. Only X_{ij} within the shaded region appear in the respective zero conditions.

we first assume that every term contains exactly $n - 3$ distinct, simple poles. In generic kinematics, a_M is extracted by taking the simultaneous residue in all poles belonging to M :

$$\text{Res}_M \mathcal{B}_n = a_M. \quad (10)$$

If the same residue remains isolated after restriction to H_T , then the zero condition immediately gives

$$0 = \text{Res}_M (\mathcal{B}_n|_{H_T}) = a_M. \quad (11)$$

The restriction can, however, force an additional channel $X_d \notin M$ to vanish together with the poles in M . A competing term M' containing X_d may then acquire the same multiple pole, so that, schematically,

$$\text{Res}_M (\mathcal{B}_n|_{H_T}) = a_M + a_{M'}. \quad (12)$$

The zero now relates two coefficients instead of eliminating a_M . We call this *pollution*. The purpose of the star test is to recognize, directly from the chords, when the pollution does not occur.

The origin of pollution is transparent in the kinematic mesh, Fig. 4. Each planar channel X_{ij} occupies a mesh vertex, while each elementary diamond carries

$$c_{ij} \equiv X_{ij} + X_{i+1,j+1} - X_{i,j+1} - X_{i+1,j} = -2p_i \cdot p_j. \quad (13)$$

An interval zero H_T sets $c_{ij} = 0$ throughout a rectangular block of diamonds, while the channels X_{ij} themselves remain generic. Summing the elementary diamond relations over any rectangle within this block gives

$$X_{ru} + X_{r'u'} = X_{ru'} + X_{r'u}, \quad r, r' \in T, \quad u, u' \notin T. \quad (14)$$

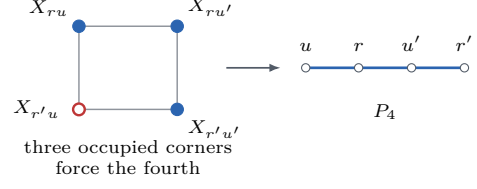


FIG. 5. The mesh origin of pollution. Three selected corners of a rectangle (shown in blue) force the fourth channel to vanish (red). Interpreting each X_{ru} as a link between r and u , the same three corners form the path P_4 .

Taking Res_M sets every pole in M to zero. If three corners of one of these rectangles are occupied by selected poles, or by the identically vanishing polygon edges bordering T , Eq. (14) forces the fourth channel to vanish as well, see Fig. 5. A second pole product can then contribute to the same residue, which is precisely the pollution described above.

This observation has a useful graphical form. For an interval T , draw every selected chord crossing between T and its complement, together with the two polygon edges at the ends of T . In Fig. 4, these chords are precisely the X_{ij} that lie in the shaded region of the particular zero. Regard each retained channel X_{ru} as a link between r and u , and call the resulting bipartite graph $\Gamma_T(S)$, where $S = \text{supp } M$. The three corners in Fig. 5 are exactly a three-link path P_4 .

A connected bipartite graph contains no P_4 precisely when all its links share one vertex, it is a star. Consequently, if every connected component of $\Gamma_T(S)$ is a star, the selected pole conditions are independent, no additional pole is forced to vanish, and the coefficient remains unpolluted. We obtain the *star test*

$$\boxed{\begin{aligned} \mathcal{B}_n|_{H_T} &= 0, \\ \Gamma_T(S) \text{ is a union of stars} \\ \implies a_M &= 0. \end{aligned}} \quad (15)$$

We call such a T a *detecting interval* for M .

At five points, for example, $M = \{13, 24\}$ and $T = \{4, 5\}$ give

$$\Gamma_T(S) = \{34, 24\} \sqcup \{51\},$$

a union of two stars, so the zero eliminates the coefficient of $1/(X_{13}X_{24})$. For the local support $\{13, 14\}$, the same interval instead produces the path $3 - 4 - 1 - 5$; the zero does not isolate this coefficient, but relates it to a neighboring triangulation. The complete extraction argument, including repeated poles, is given in Appendix 1.

The locality problem is now graphical. For every non-triangulating support, we must find at least one interval whose cut graph is a union of stars. We construct such an interval recursively below.

FINDING THE ZERO RECURSIVELY

The star test verifies a proposed zero; it remains to find one. Starting from a nontriangulating support S_n , we remove a suitable vertex p , apply the result recursively to the resulting $(n-1)$ -gon, and lift the child zero interval back to the original polygon,

$$S_n \longrightarrow S_{n-1} \longrightarrow T_{n-1} \longrightarrow T_n. \quad (16)$$

Once a detecting interval T_{n-1} has been found, there are only two possible lifts: either

$$T_n = T_{n-1}, \quad \text{or} \quad T_n = T_{n-1} \cup \{p\}. \quad (17)$$

Thus the only question is whether p must be added to the child zero interval. Remarkably, the answer depends only on the local chord configuration around p . The counting result below guarantees that a suitable p always exists, reducing the all-multiplicity construction to four local cases.

The recursion rests on a simple but crucial counting fact. If a free vertex exists, meaning that no selected chord ends there, we choose it. Otherwise, let L be the number of leaves, namely vertices touched by exactly one selected chord. Every other vertex is touched at least twice, and therefore

$$2|S_n| \geq L + 2(n - L). \quad (18)$$

Since $|S_n| \leq n - 3$, it follows that

$$\boxed{\text{no free vertex} \implies L \geq 6.} \quad (19)$$

The result is essential. At full chord count, a nontriangulating support contains a crossing pair. Protecting its four endpoints leaves at least two leaves available for removal, so the crossing survives in the child. Below full chord count, removing any leaf preserves the chord deficit. Thus the child is nontriangulating in either case. This simple counting result is the key to the recursion: it guarantees a removable vertex and reduces the all-multiplicity proof to the four local configurations below.

Let a, p, b be consecutive vertices and let X_{ab} be the short chord above p . The chosen vertex is either free or a leaf, and X_{ab} is either absent or present. There are therefore only four cases, shown in Fig. 3. In F0 the zero interval $\{a, p\}$ detects the term directly. In F1 one deletes p and X_{ab} , then lifts a detecting zero of the child. In L0 one either finds a nearby detecting zero directly or deletes p and its unique incident chord before recursing. In L1 the short and the leaf chord are removed, and the leaf chord is reattached through a or through b , producing two possible children.

The lifting lemma states that, in F1 and L0, every detecting zero interval of the prescribed child has a allowed lift that detects the parent. In L1, choose a detecting

interval from each of the two child configurations. Either interval may fail to lift because restoring p creates a P_4 , but the two-slide lemma guarantees that they cannot both fail, at least one has a allowed detecting lift to the parent. Table I exhausts the local lifting possibilities.

Thus every branch either produces a detecting zero directly or reduces the problem to lower multiplicity. The complete lifting rules, the proof of the two-slide lemma, worked examples, and higher-multiplicity checks are given in Appendix 2. Appendix 2 also proves that triangulations evade every star test.

The construction is therefore directly executable. Given any excluded pole product, we deterministically obtain a hidden-zero interval that isolates its coefficient. A Wolfram Language implementation that constructs the child configurations, tests their zero intervals, and displays every attempted lift and its star-test certificate is included as ancillary material. The lifting lemmas were also exhaustively checked through $n = 12$; the corresponding case counts are given in Table II.

OUTLOOK

This result closes an all-multiplicity question about emergent locality first raised in [4, 10]. More strongly than a uniqueness theorem, it provides a direct method for extracting local graph structure from regular kinematic zeros.

For local ansätze, hidden zeros are equivalent to dual shuffle factorization into D -subsets [27, 32]. Without assuming locality, our result suggests that this factorization is the local limit of a more general structure in which zeros organize both local and nonlocal pole products into related classes.

The immediate extensions are the NLSM and Yang–Mills theory, which are organized by the same planar variables and possess related hidden zeros [19, 23, 33–36]. At loop level, the surface framework represents planar diagrams by curves on punctured disks and develops all-loop curve and cut-equation formulations [28, 29, 37–42]. The corresponding surface integrands obey “big-mountain” zeros, which already give low-point evidence for emergent loop-level locality [11]. These extensions are currently ongoing.

The theorem has a purely combinatorial reading that may be of independent interest. Triangulations of a polygon are usually characterized by what they contain, $n-3$ pairwise noncrossing chords, or by indexing the vertices of the associahedron [30], equivalently the facets of the type- A cluster complex [43]. Here they are characterized by what they evade. Among configurations of at most $n-3$ chords, triangulations are exactly those for which no interval cut graph is a union of stars. We are not aware of this characterization in the combinatorics literature.

More broadly, the proof demonstrates the power of

TABLE I. Complete lifting rules.

Let $T = T_{n-1}$ and $T^+ = T \cup \{p\}$. In the last column, assume wlog $a \in T$ and $b \notin T$. $\Gamma(S)$ denotes the cut graph.			
Case	Condition/reduction	a, b on the same side	a, b on opposite sides
F0	p free, $X_{ab} \notin S_n$		$T_n = \{a, p\}$
F1	p free, $X_{ab} \in S_n$, $S_{n-1} = S_n \setminus \{X_{ab}\}$	$a, b \in T : T_n = T^+$, $a, b \notin T : T_n = T$	$\nexists X_{bv} \in \Gamma_T(S_{n-1}), v \neq a : T_n = T$, $\nexists X_{av} \in \Gamma_T(S_{n-1}), v \neq b : T_n = T^+$
L0	$X_{au} \notin \Gamma(S_n)$		$T_n = \{a, p\}$
	$X_{bu} \notin \Gamma(S_n)$		$T_n = \{p, b\}$
	$X_{au}, X_{bu} \in \Gamma(S_n)$, $S_{n-1} = S_n \setminus \{X_{pu}\}$	$a, b \in T : T_n = T^+$, $a, b \notin T : T_n = T$	$u \in T : T_n = T^+$, $u \notin T : T_n = T$
L1	$R = S_n \setminus \{X_{pu}, X_{ab}\}$, $S_a = R \cup \{X_{au}\}$, $S_b = R \cup \{X_{bu}\}$	$a, b \in T : T_n = T^+$, $a, b \notin T : T_n = T$	$u \in T : \text{first try } T^+ \text{ then if it fails } T$, $u \notin T : \text{first try } T \text{ then if it fails } T^+$
Apply first to S_a . If neither lift works, repeat with S_b ; the two-slide lemma guarantees success.			

surfaceology. It turns a question that is elementary to state, yet has resisted conventional on-shell methods, into a finite combinatorial problem. In doing so, it reveals structures in scattering amplitudes that become visible only through surface geometry and regular kinematic zeros.

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APPENDIX 1: PROOF OF THE STAR TEST

We prove the star test for the full ansatz (6), including repeated poles and terms with fewer than $n - 3$ denominator factors. Fix a zero interval

$$T = \{w, w + 1, \dots, w + k\}, \quad v = w + k + 1, \quad 2 \leq |T| \leq n - 3, \quad (1.1)$$

and let T^c denote its complement. For a pole multiset M , let $S = \text{supp } M$ be the set of distinct chords appearing in M . A chord belonging to S will be called selected. The cut graph $\Gamma_T(S)$ contains the selected chords crossing between T and T^c , together with the two polygon edges at the ends of T . We prove

$$\Gamma_T(S) \text{ is a union of stars,} \quad \mathcal{B}_n|_{H_T} = 0 \quad \implies \quad a_M = 0. \quad (1.2)$$

We work in generic formal kinematics, without dimension-dependent Gram relations.

Coordinates on the zero.— On H_T , every channel crossing the cut can be written as

$$X_{ru} = A_r + B_u, \quad r \in T, \quad u \in T^c. \quad (1.3)$$

The two polygon edges at the ends of T impose

$$A_w + B_{w-1} = 0, \quad A_{w+k} + B_v = 0. \quad (1.4)$$

Explicitly, one may take

$$A_r = X_{r,v}, \quad B_u = X_{w,u} - X_{w,v}, \quad (1.5)$$

which solves the zero relations (8). Equation (1.3) then gives

$$X_{ru} + X_{r'u'} = X_{ru'} + X_{r'u}, \quad (1.6)$$

which is the rectangle relation used in the main text. Channels whose endpoints lie on the same side of the cut remain independent.

Graph criterion.— Temporarily set to zero every link of $\Gamma_T(S)$, including its two boundary links. Writing $C_u = -B_u$, these conditions become $A_r = C_u$. They are independent precisely when the cut graph is a forest. Moreover, they force an additional channel X_{ru} to vanish precisely when r and u lie in the same connected component.

It follows that no unselected channel is forced to vanish exactly when each connected component contains every possible link between its two sets of vertices, that is, when each component is complete bipartite. A connected complete bipartite graph is also a tree precisely when one side contains a single vertex. Such a graph is a star. Therefore

$$\begin{aligned} &\Gamma_T(S) \text{ is a union of stars} \\ &\iff \\ &\text{the selected pole conditions are independent and} \\ &\text{force no additional channel to vanish.} \end{aligned}$$

(1.7)

This is also the precise form of the mesh argument: three occupied corners of a rectangle form a P_4 and force the fourth corner to vanish.

Isolation of the coefficient.— Let C_α be the connected components of $\Gamma_T(S)$, including isolated vertices, and choose distinct constants λ_α . Introduce a large parameter Λ and set

$$A_r = \lambda_\alpha \Lambda + a_r, \quad B_u = -\lambda_\alpha \Lambda + b_u, \quad r, u \in C_\alpha. \quad (1.8)$$

This deformation remains entirely inside H_T . Because each nontrivial component is a star, its finite parameters may be chosen so that all selected channels crossing the cut take arbitrary independent values. We also assign independent variables to all channels lying entirely on one side of the cut.

Every selected channel remains finite. By Eq. (1.7), an unselected channel crossing the cut joins two different components and therefore grows as

$$X_{ru} = (\lambda_\alpha - \lambda_\beta) \Lambda + a_r + b_u = O(\Lambda), \quad \alpha \neq \beta. \quad (1.9)$$

Since the numerators in (6) are constant, every term containing such a channel vanishes as $\Lambda \rightarrow \infty$.

The restricted zero condition consequently becomes a Laurent identity in the independent finite channels. Distinct pole multisets give distinct Laurent monomials, so the coefficient of each monomial must vanish separately. In particular, the monomial with precisely the pole multiplicities of M has coefficient a_M , and hence $a_M = 0$.

Repeated poles are distinguished by their Laurent exponents, while terms with fewer factors have different zero exponents. Thus the same argument covers the full ansatz. The bound $|M| \leq n - 3$ is not required for the star test itself; it enters only in Appendix 2, where it guarantees the existence of a detecting interval.

APPENDIX 2: RECURSIVE CONSTRUCTION

We now prove that every nontriangulating support within the chord bound has a detecting interval. For the cut-graph problem, T and T^c define the same partition; we always choose the representative satisfying $2 \leq |T| \leq n - 3$. A zero interval is detecting when every connected component of its cut graph is a star. We prove

$$|S| \leq n - 3, \quad S \text{ is not a triangulation} \implies \exists T : \Gamma_T(S) \text{ is a union of stars.} \quad (2.1)$$

Pole multiplicities play no role here because the cut graph depends only on the support S .

Induction and allowed lifts

There are two induction branches. In the *deficit branch*, $|S| \leq n - 4$. In the *protected-crossing branch*, $|S| \leq n - 3$ and S contains a chosen crossing pair whose four endpoints are protected throughout the reduction. Every support in Eq. (2.1) belongs to at least one branch. If $|S| = n - 3$, a nontriangulating support must contain a crossing pair.

Let a, p, b be consecutive vertices of the current polygon. After deleting p , the vertices a and b become adjacent; we call ab the new child boundary edge. A parent interval T_n lifts a child interval T_{n-1} if deleting p from T_n gives T_{n-1} . There are at most two possibilities:

$$T_n = T_{n-1}, \quad \text{or} \quad T_n = T_{n-1} \cup \{p\}. \quad (2.2)$$

Only choices that remain cyclic intervals are allowed.

If $a, b \in T_{n-1}$, the unique lift is $T_n = T_{n-1} \cup \{p\}$. If $a, b \notin T_{n-1}$, the unique lift is $T_n = T_{n-1}$. If exactly one of a, b belongs to T_{n-1} , both choices are possible.

The induction begins at five points. At full chord count, every nontriangulating two-chord support is a crossing pair. Its fifth vertex is free, and the short chord above that vertex is absent, so the F0 construction below applies. Supports with at most one chord are treated in the same way after choosing a free vertex whose short chord is absent.

We next show that a suitable vertex always exists. A vertex is *free* if no selected chord ends there and is a *leaf* if exactly one selected chord ends there. If there is no free vertex and L vertices are leaves, then

$$2|S| = \sum_v \deg_S(v) \geq L + 2(n - L). \quad (2.3)$$

Since $|S| \leq n - 3$,

$$L \geq 6. \quad (2.4)$$

In the deficit branch, deleting any leaf and its unique chord leaves at most $n - 5$ chords, so the child still has a chord deficit. In the protected-crossing branch, choose a leaf away from the four protected endpoints; the six-leaf bound guarantees at least two choices. Deleting this leaf preserves the protected crossing.

The short chord X_{ab} is never part of the protected pair. Indeed, any chord crossing X_{ab} must end at p , because X_{ab} cuts off the triangle apb . Since p is not a protected endpoint, X_{ab} cannot be protected. Thus every child constructed below satisfies the same inductive condition as its parent.

The four cases are determined by whether p is free or a leaf and whether the short chord X_{ab} is absent or present.

F0: Free vertex, short absent

Suppose p is free and $X_{ab} \notin S$. Then

$$T_n = \{a, p\} \quad (2.5)$$

is detecting. Every selected chord crossing this interval ends at a , since none ends at p . These chords and the polygon edge immediately preceding a therefore form a star centered at a . The boundary edge pb is a separate one-edge star, and the absence of X_{ab} prevents the two components from joining.

This is the direct five-point construction illustrated in the main text, so no additional figure is needed here.

F1: Free vertex, short present

Suppose p is free and $X_{ab} \in S$. Delete p and X_{ab} , giving

$$S_{n-1} = S \setminus \{X_{ab}\}. \quad (2.6)$$

Let T_{n-1} be any detecting interval of the child.

If $a, b \in T_{n-1}$, choose $T_n = T_{n-1} \cup \{p\}$. If $a, b \notin T_{n-1}$, choose $T_n = T_{n-1}$. In either case a, p, b all lie on the same side of the parent cut, so the restored chord X_{ab} does not appear in the parent cut graph, which is unchanged.

Now suppose $a \in T_{n-1}$ and $b \notin T_{n-1}$. Then ab is a boundary link of the child cut graph. Because that graph is a union of stars, other links crossing the cut can end at a or at b , but not at both, otherwise they and ab would form a P_4 .

If the other links end at a , choose $T_n = T_{n-1}$. The child boundary link ab is then replaced by the parent boundary link ap and the restored chord X_{ab} , both incident on a . If the other links end at b , choose $T_n = T_{n-1} \cup \{p\}$; the corresponding two links are now pb and X_{ab} , both incident on b . If neither endpoint has another link, either lift works. Hence every detecting interval of the child has an allowed detecting lift.

Example. For $S_6 = \{13, 15, 24\}$, choose the free vertex $p = 6$. Deleting 6 and the short chord 15 gives $S_5 = \{13, 24\}$. The detecting interval $T_5 = \{4, 5\}$ crosses the new child edge 51. Choosing $T_6 = T_5$ replaces the child link 51 by the two links 56 and 15, both centered at 5.

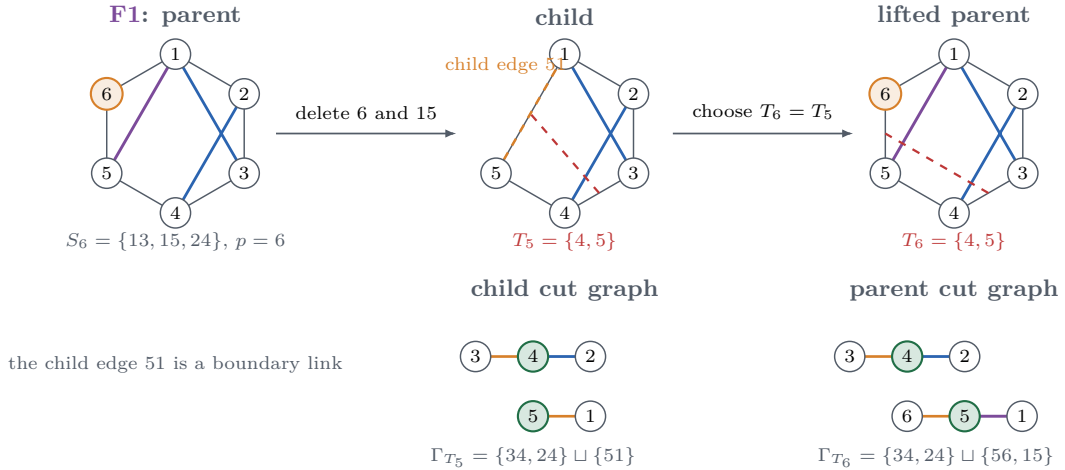


FIG. 6. F1 when the new child edge 51 is a boundary link of the child cut graph. Choosing $T_6 = T_5 = \{4, 5\}$ leaves both 6 and 1 outside the interval. The one-edge star 51 becomes the two-arm star $6 - 5 - 1$.

L0: Leaf, short absent

Let X_{pu} be the unique selected chord ending at p , and suppose $X_{ab} \notin S$.

If X_{au} is neither a selected chord nor a polygon edge, then $T_n = \{a, p\}$ is detecting. The links ending at a form one star, while X_{pu} and the boundary edge pb form another star centered at p . The two components cannot join through u , because X_{au} is absent, or through b , because X_{ab} is absent. If instead X_{bu} is neither a selected chord nor a polygon edge, the reflected interval $T_n = \{p, b\}$ is detecting.

It remains to consider the case in which both X_{au} and X_{bu} are either selected chords or polygon edges. Delete p and X_{pu} , giving

$$S_{n-1} = S \setminus \{X_{pu}\}, \quad (2.7)$$

and let T_{n-1} be any detecting interval of the child.

First suppose a and b lie on the same side of the child cut. Use the unique lift in Eq. (2.2). If p and u lie on the same side of the parent cut, X_{pu} does not appear in its cut graph. Otherwise both X_{au} and X_{bu} cross the child cut. Since the child graph is a union of stars, their component must be centered at u . Restoring X_{pu} then adds one more arm to that same star.

Now suppose $a \in T_{n-1}$ and $b \notin T_{n-1}$. If $u \notin T_{n-1}$, choose $T_n = T_{n-1}$. Then p and u are both outside T_n , so X_{pu} does not cross the parent cut. In the child, the links X_{au} and ab force their component to be centered at a ; replacing the boundary link ab by the new boundary link ap preserves that star. If $u \in T_{n-1}$, choose $T_n = T_{n-1} \cup \{p\}$. Now X_{pu} again stays on one side, while X_{bu} and ab force the child star to be centered at b ; replacing ab by pb preserves it. Thus every detecting child interval again has an allowed detecting lift.

Direct example. For $S_6 = \{14, 25, 36\}$, choose the leaf $p = 6$, whose unique chord is 36. The short chord 15 and the link 35 are absent, so $T_6 = \{5, 6\}$ detects the parent directly:

$$\Gamma_{T_6}(S_6) = \{45, 25\} \sqcup \{61, 36\}.$$

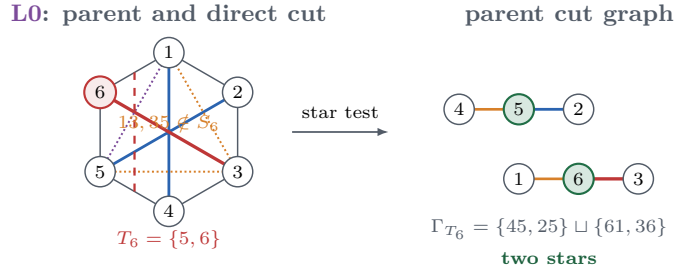


FIG. 7. The direct L0 move. The missing link 35 makes the displayed two-vertex interval detecting; the reflected construction uses the missing link 13.

Recursive example. For $S_6 = \{13, 24, 36\}$, choose $p = 1$, with unique chord 13. Deleting 1 and 13 gives $S_5 = \{24, 36\}$. The interval $T_5 = \{5, 6\}$ crosses the new child edge 62. Since $u = 3$ lies outside T_5 , choose $T_6 = T_5$, so $p = 1$ also lies outside the parent interval. The restored chord 13 is then absent from the cut graph, while the child boundary link 62 is replaced by 61.

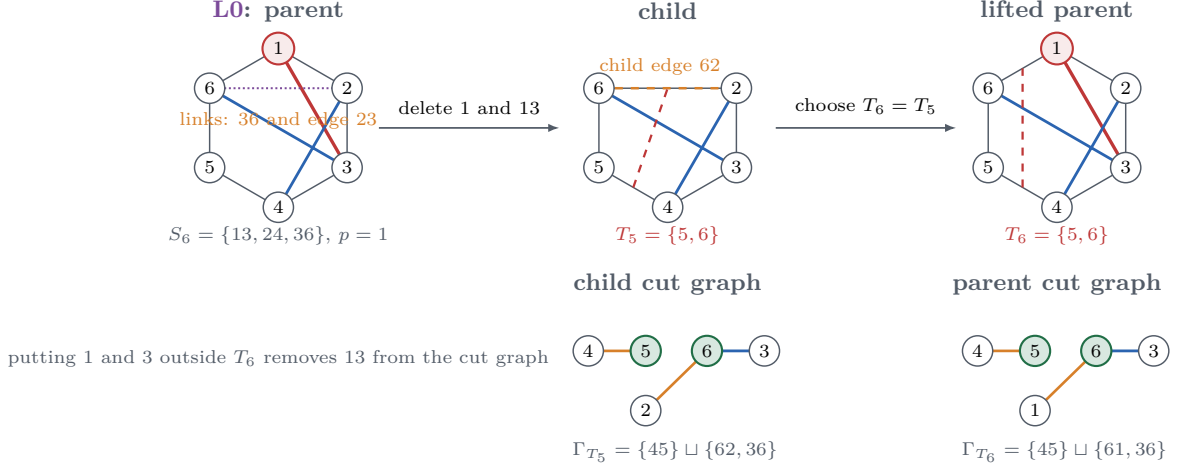


FIG. 8. L0 when the new child edge 62 is a boundary link of the child cut graph. Choosing $T_6 = T_5 = \{5, 6\}$ places both $p = 1$ and $u = 3$ outside the interval, so 13 does not cross the cut. The boundary link 62 is replaced by 61.

L1: Leaf, short present

Finally suppose $X_{pu}, X_{ab} \in S$. Delete p , X_{pu} , and X_{ab} , and write

$$R = S \setminus \{X_{pu}, X_{ab}\}. \quad (2.8)$$

Form two child supports

$$S_a = R \cup \{X_{au}\}, \quad S_b = R \cup \{X_{bu}\}. \quad (2.9)$$

We call these the a -child and b -child. If the added link is a polygon edge, it is omitted from the support; if it already belongs to R , it is not duplicated. Both children therefore contain at most $n - 4$ selected chords and preserve the relevant induction condition.

Choose arbitrary detecting intervals T_a and T_b for the two children. To reconstruct the parent, restore p , X_{pu} , and X_{ab} . A link X_{zu} , $z = a, b$, that was introduced only in the child is removed; if it already belonged to R , it remains selected in the parent.

A detecting interval of one child need not lift. For example, suppose the a -child contains an additional selected link X_{ay} , so its cut graph contains the star $y - a - u$. If $X_{au} \in R$, this link remains in the parent. Restoring X_{pu} then creates the path $y - a - u - p$, which is a P_4 . The role of the second child is to avoid this obstruction.

Two-slide lemma. *For any detecting intervals T_a and T_b of the two children, at least one has an allowed lift for which the parent cut graph is a union of stars.*

Proof. For $z \in \{a, b\}$, let $\bar{z}$ denote the other vertex in $\{a, b\}$. Call T_z obstructed for the z -child if none of its allowed parent lifts is detecting.

First suppose a and b lie on the same side of T_z , so the lift is unique. If u lies on that same side, neither restored chord crosses the parent cut and the lift succeeds. If u lies on the opposite side and X_{zu} was introduced only in the child, the lift simply replaces the child link X_{zu} by X_{pu} . If the child star is centered at u , one arm is exchanged for another; if it is centered at z , removing X_{zu} leaves X_{pu} as a separate one-edge star. The lift again succeeds.

An obstruction with a and b on the same side is therefore possible only when X_{zu} remains in the parent, either because

$$\begin{aligned} (\text{I}_s) \quad & X_{zu} \in R, \\ (\text{I}_e) \quad & X_{zu} \text{ is a polygon edge.} \end{aligned} \quad (2.10)$$

In either case, the child component containing X_{zu} must be centered at z and must contain a second link X_{zy_z} , where y_z denotes its other endpoint. Restoring X_{pu} then creates $y_z - z - u - p$. In case (I_e) , the second link X_{zy_z} must be selected; if it were also a boundary link, one side of the zero interval would contain only z , which is not allowed.

Now suppose ab is a boundary link of the child cut graph. If u lies on the same side as $\bar{z}$, choose the lift that places p on that side. Then X_{pu} does not cross the parent cut. The restored short X_{ab} and the new parent boundary link both end at z . The child link X_{zu} , together with the boundary link ab , already forces the child component to be centered at z , so this replacement preserves a star.

An obstructed interval must therefore place u on the same side as z . Testing the other lift shows that failure is possible only if

$$(B) \quad ab \text{ belongs to a star centered at } z \text{ containing another selected link } X_{zyz}. \quad (2.11)$$

The parent then contains the path $y_z - z - \bar{z} - p$. Thus every obstructed z -interval separates $\bar{z}$ from u .

Assume, for contradiction, that both T_a and T_b are obstructed. The pair (I_e, I_e) is impossible because u cannot be a polygon neighbor of both consecutive vertices a and b for $n \geq 5$. The pairs (I_e, B) and (B, I_e) would leave one side of one interval with only one vertex and are therefore also impossible.

Every remaining pairing involving an internal obstruction creates a P_4 in one of the two child cut graphs:

type of T_a	type of T_b	forbidden child path
I_s	I_s or I_e	$a - u - b - y_b \subset \Gamma_{T_b}(S_b)$
I_e	I_s	$b - u - a - y_a \subset \Gamma_{T_a}(S_a)$
I_s	B	$u - a - b - y_b \subset \Gamma_{T_b}(S_b)$
B	I_s	$u - b - a - y_a \subset \Gamma_{T_a}(S_a)$

For instance, in the first row the obstruction for T_a implies $X_{au} \in R$. Since an obstructed b -interval separates a from u , this chord crosses the b -cut. The obstruction for T_b supplies X_{ub} and X_{by_b} , producing the displayed P_4 . The other rows follow identically or by reflection.

It remains to exclude (B, B) . For an interval whose cut graph contains the boundary link ab , let B_z denote the side of T_z containing b . Such sides are cyclic intervals beginning at b , and hence B_a and B_b are nested. The obstruction for T_a gives $u \notin B_a$, whereas the obstruction for T_b gives $u \in B_b$; therefore $B_a \subsetneq B_b$. It also gives $y_a \in B_a$, while $a \notin B_b$, so X_{ay_a} crosses the b -cut. Together with X_{ab} and X_{by_b} , it forms

$$y_a - a - b - y_b \subset \Gamma_{T_b}(S_b),$$

contradicting the assumption that T_b is detecting.

Thus T_a and T_b cannot both be obstructed. At least one has an allowed detecting lift to the parent. $\square$

Example. Let

$$S_7 = \{14, 16, 27, 35\}, \quad p = 7, \quad a = 6, \quad b = 1, \quad u = 2.$$

After deleting 7, 27, 16, the b -child is $S_b = \{14, 35\}$, since 12 is a polygon edge. Its detecting interval $T_b = \{1, 2\}$ has two allowed parent lifts. The lift $T_7 = \{1, 2\}$ contains the P_4 , $4 - 1 - 7 - 2$, while the lift $T_7 = \{7, 1, 2\}$ contains the P_4 , $7 - 6 - 1 - 4$; thus neither lift is detecting. The a -child is $S_a = \{14, 26, 35\}$. Its interval $T_a = \{2, 3\}$ lifts with $T_7 = T_a$, replacing the arm 26 by 27:

$$\Gamma_{T_a}(S_a) = \{12, 26\} \sqcup \{34, 35\}, \quad \Gamma_{T_7}(S_7) = \{12, 27\} \sqcup \{34, 35\}.$$

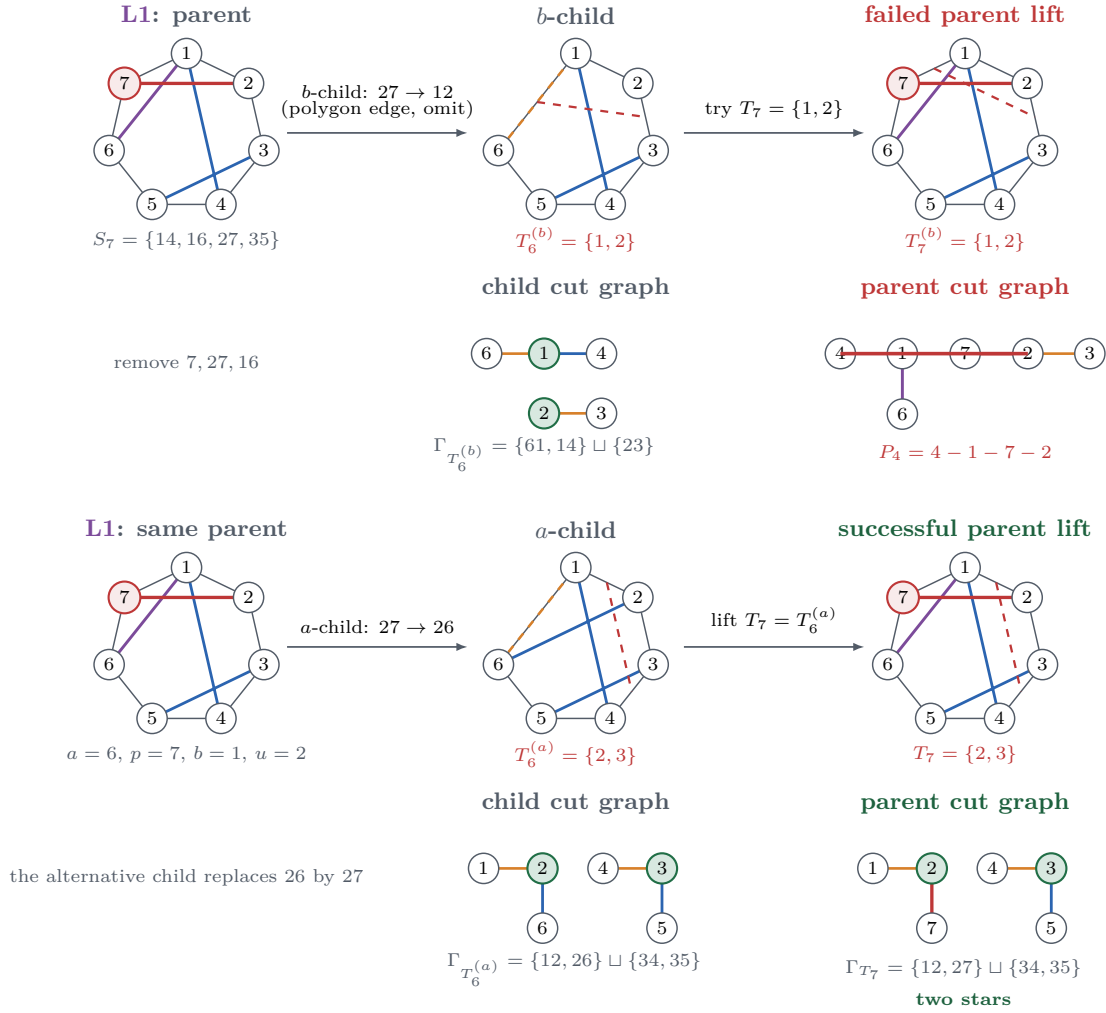


FIG. 9. The two children in L1. The upper row displays the obstructed lift $T_7 = \{1, 2\}$ from the b -child; the other allowed lift, $T_7 = \{7, 1, 2\}$, is also obstructed, by $7 - 6 - 1 - 4$. The a -child instead supplies a detecting lift. It replaces the arm 26 by 27, so the parent cut graph remains a union of stars.

Completion

The four cases are exhaustive because the chosen vertex is free or a leaf and its short chord is absent or present. F0 returns a detecting interval directly. In F1, every detecting interval of the prescribed child has an allowed detecting lift. In L0, either one of the two-vertex intervals works directly or every detecting interval of the prescribed child lifts. In L1, the two-slide lemma guarantees that detecting intervals from the two children cannot both be obstructed. Every recursive step lowers the number of vertices, so the construction reaches the five-point base case and proves Eq. (2.1).

The proof is constructive. Fix an ordering of vertices, crossing pairs, zero intervals, and allowed lifts. At each stage choose the first free vertex; if none exists, choose the first admissible leaf. Apply the appropriate rule above and recurse. In L1, obtain one detecting interval from each child and test their allowed lifts in the fixed order. The two-slide lemma guarantees that this finite procedure returns a detecting zero interval.

Why triangulations evade every test

Let S triangulate the polygon and let T be any allowed zero interval. Color the vertices in T and T^c by 0 and 1, respectively, and extend the coloring linearly across every triangle. The level set $f = \frac{1}{2}$ meets the polygon boundary at exactly the two boundary links of the cut. Inside each triangle it is a line segment joining two sides. Because the dual

graph of a polygon triangulation is a tree, these segments cannot form a closed component. They therefore form one arc joining the two boundary links.

Consecutive links crossed by this arc belong to a common triangle and hence share a vertex. The two boundary links consequently lie in one connected component of $\Gamma_T(S)$. They are disjoint because both sides of the zero interval contain at least two vertices, so that component cannot be a star. No zero interval detects a triangulation.

Together with Eq. (2.1), this proves that, among supports containing at most $n - 3$ chords, triangulations are precisely those that evade every interval star test.

Explicit checks

We counted exactly through twelve points all nonlocal pole multisets and all admissible choices of the removed vertex. A rooted recursion case is a pair (M, p) , where p is a free vertex or, if none exists, a leaf whose deletion preserves the relevant induction condition. At full support size, this means that at least one crossing pair survives the deletion.

The recursion depends only on $S = \text{supp } M$. Nevertheless, different multiplicities are counted as different terms. A support of size s has $\binom{n-3}{s}$ positive multiplicity assignments of total degree at most $n - 3$.

We also tested every detecting child interval and every allowed lift, finding no counterexample to the lifting lemmas. Individual L1 lifts can be obstructed, but no simultaneous two-child obstruction was found. OpenAI Codex was used to assist with implementing and auditing these explicit computer checks.

TABLE II. Exact numbers of nonlocal pole multisets M and admissible recursion choices (M, p) . Repeated poles and terms with fewer than $n - 3$ factors are included. A pole multiset may contribute more than once because several vertices p may be admissible.

n	N_M	F0	F1	L0	L1
5	16	30	10	0	0
6	206	336	138	18	0
7	3,018	5,005	1,904	308	84
8	52,998	93,024	30,672	6,360	1,872
9	1,107,139	2,072,070	590,832	142,614	37,728
10	26,976,898	53,796,160	13,444,750	3,535,490	827,290
11	752,533,288	1,595,093,643	354,449,524	96,783,192	20,231,948
12	23,667,673,019	53,179,362,600	10,635,814,176	2,912,891,436	550,246,848